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AWS + Linux Interview Guide

For 3+ Years Experience

Structured Q&A Format with Real Interview Approach

Each section below includes frequently asked questions, how to answer in interviews, and tips to tackle them.

1. What is Cloud Computing?

Q: What is cloud computing? **A:** It's accessing computing services over the internet (like servers, storage, databases) instead of local computers.

Q: Types of cloud? **A:** Public, Private, and Hybrid Cloud.

Q: What are IaaS, PaaS, SaaS? **A:** Service models:

- IaaS: Infrastructure (EC2)
- PaaS: Platform (Elastic Beanstalk)
- SaaS: Software (Gmail)

How to answer: Give clear real examples from AWS (e.g., EC2 = IaaS).

2. EC2 (Elastic Compute Cloud)

Q: What is EC2? **A:** EC2 is a virtual server in AWS.

Q: What are EC2 instance types? **A:** General (t2), Compute (c5), Memory (r5), Storage (i3).

Q: What are EC2 pricing models? **A:** On-demand, Spot, Reserved, and Savings Plans.

Q: How can you launch EC2? **A:** Console, CLI, CloudFormation, or Launch Template.

Tips: Always mention cost-saving with Spot and Reserved Instances.

3. S3 (Simple Storage Service)

Q: What is S3? **A:** S3 is object storage used to store and retrieve any amount of data.

Q: What is S3 versioning? **A:** Keeps all versions of a file even if deleted.

Q: How to save cost? **A:** Use Lifecycle Policy to move to Glacier.

Q: How to upload via CLI? **A:** `aws s3 cp file.txt s3://bucket/`

Scenario: You need a website with no server. Use S3 static hosting.

4. CloudWatch

Q: What is CloudWatch? **A:** Monitors resources like EC2, stores logs, and triggers alarms.

Q: How to monitor high CPU? **A:** Set CloudWatch alarm and send notification via SNS.

Q: What is Logs Insights? **A:** Use SQL-like queries to search logs.

5. CloudTrail

Q: What is CloudTrail? **A:** Tracks user activity and API usage in AWS.

Q: Where are logs stored? **A:** In S3 bucket.

Tips: Mention use for auditing and security reviews.

6. EFS

Q: What is EFS? **A:** A shared file system for EC2 using NFS protocol.

Q: Can multiple EC2s use it? **A:** Yes, across AZs.

Scenario: You need common storage between app servers → Use EFS.

7. Monitoring & Automation

Q: How to monitor EC2? **A:** CloudWatch for metrics/logs, Alarm + SNS for alerts.

Q: How to automate backup? **A:** Use CloudWatch Event Rule + Lambda or Lifecycle Policy.

8. Linux Essentials

Q: How to find error lines? **A:** `grep 'error' logfile.log`

Q: How to copy folder? **A:** `cp -r source/ destination/`

Q: Difference between hard and soft link?

Hard Link	Soft Link
Same inode	Pointer to path
Not broken if source deleted	Broken if source deleted

9. Protocols

Q: Which protocol is used for SSH? **A:** TCP port 22.

Q: Which port for HTTPS? **A:** 443.

10. Security Groups

Q: What are Security Groups? **A:** Virtual firewalls for EC2 that control inbound and outbound traffic.

Q: Can you block traffic in SG? **A:** No, only allow rules are supported.

11. VPC

Q: What is VPC? **A:** A private network inside AWS.

Q: Difference between Public and Private Subnet? **A:** Public has IGW; Private uses NAT Gateway.

Scenario: Database should not be public — put in private subnet.

12. Load Balancers

Q: What is ALB? **A:** Application Load Balancer for HTTP/HTTPS (Layer 7).

Q: What is NLB? **A:** Network Load Balancer for TCP traffic (Layer 4).

Q: What is Target Group? **A:** A group of EC2s behind the Load Balancer.

13. Auto Scaling Group

Q: Why use ASG? **A:** Automatically adds/removes EC2 based on demand.

Q: What triggers scaling? **A:** CloudWatch Alarms (e.g., CPU > 80%).

14. CloudFormation

Q: What is CloudFormation? **A:** It helps deploy resources via templates (YAML/JSON).

Tips: Use for IaC (Infrastructure as Code).

15. AWS CLI

Q: What is AWS CLI? **A:** Command-line tool to manage AWS resources.

Q: Example to list buckets? **A:** `aws s3 ls`

Q: Launch EC2?

```
aws ec2 run-instances --image-id ami-xxxx --instance-type t2.micro
```

16. IAM Roles & Policies

Q: What is IAM Role? **A:** Permission assigned to AWS services (e.g., EC2 access to S3).

Q: What is IAM Policy? **A:** JSON file that defines allowed actions.

17. NACL vs Security Group

Feature	SG	NACL
Scope	Instance	Subnet
Stateful	Yes	No

Feature	SG	NACL
Allow/Deny	Only Allow	Allow + Deny

18. EC2 Metadata

Q: How to get EC2 metadata?

```
curl http://169.254.169.254/latest/meta-data/
```

Use: Find public IP, instance ID, IAM Role from EC2 itself.

19. VPC Peering

Q: What is VPC Peering? **A:** Connects 2 VPCs so resources can talk.

Q: Do you need route tables updated? **A:** Yes, to allow traffic.

20. Lifecycle Management in S3

Q: Why use it? **A:** Save cost by moving old files to Glacier or delete them.

Q: Example Rule? **A:** Move to Glacier after 30 days, delete after 180 days.

21. CloudWatch Logs Insights

Q: How to search error logs?

```
fields @timestamp, @message
| filter @message like /error/
```

Use: Analyze logs in real-time.

22. Scenario-Based Quick Q&A

Q: How to send alert when new file uploaded to S3? **A:** S3 Event → EventBridge → SNS → Email.

Q: EC2s need shared storage? **A:** Use EFS.

Q: Backend EC2 should not access internet? **A:** Put in private subnet.

Q: Deploy WordPress with scaling? **A:** EC2 + ALB + ASG + RDS.

Q: Track who terminated an EC2? **A:** Use CloudTrail.

End of Guide - With Q&A and Interview Handling Tips